

Robert Tylman

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EXPERIENCE

Camp Solelim — Assistant Director & Head Tripper

Sudbury, Canada · May 2021 – Aug 2025

- Developed an internal logistics and route-planning platform leveraging Google Earth APIs for campsite optimization, meal planning, and route visualization, reducing seasonal costs by \$10K+ and increasing trip enrollment by 500%.
- Designed and maintained *CampSorter*, a full-stack scheduling and logistics platform (JavaScript, Python, SQL), reducing admin workload by 75% and improving camp programming efficiency; system remains in active use.

People's Trust Company — Software Engineer & IT Analyst Intern

Toronto, Canada · May 2023 – Aug 2023

- Partnered with finance and operations teams to audit their manual data workflows, then built Python automation pipelines that replaced the most error-prone steps and cut input errors by 95%.
- Owned company-wide IT infrastructure for 600+ employees: encryption, networking, onboarding, permissions, security.

OurCrowd/Labs02 — Data Science/SWE Intern

Jerusalem, Israel · Jan 2021 – May 2021

- Engineered an automated milestone-tracking dashboard for 5–7 portfolio startups (Google Sheets + Python + Google Calendar API), syncing fundraising and sprint deadlines; adopted as standard practice after my internship.

MyPart — Machine Learning Intern

Tel Aviv, Israel · Sep 2020 – May 2021

- Refined retrieval and feature-extraction pipelines over a 250,000+ track dataset, then trained multimodal embedding models for recommendation and semantic similarity ranking, improving retrieval relevance by 15% over baseline.

PROJECTS

RAG Research Extension — Chrome Extension (MV3), Ragas, React, Tavily API, TypeScript

- Built a lightweight Chrome extension for research and article fact-checking using Tavily's */extract* and */research* API, adding async polling and result parsing to deliver source-supported analysis on user-selected text.
- Engineered a provider-agnostic RAG evaluation pipeline benchmarking retrieval quality, latency, citation grounding, and Ragas metrics (faithfulness, answer relevance, context precision) across Tavily, Exa, Parallel, Brave, and Firecrawl.

Ableton MCP Server — Ableton Live API, Agents, MCP, Python, Socket/UDP

- Collaborated on a production MCP server exposing Ableton Live as a tool surface for LLM agents (Claude, Codex); designed the tool schema and state layer so natural-language prompts translate into real-time software operations (track management, volume automation, MIDI programming, device manipulation).

Twist! — Expo, Gemini 3.1 API, Node.js, React, Redis, Socket.io

- Shipped a production multiplayer party-game app on a 4-developer team where users vote on AI-generated image transformations of their own photos; reached 350+ players and \$100+ revenue in the first month.
- Designed disconnection recovery using heartbeat-based liveness detection and cross-state server synchronization.

Multimodal Audio-to-Text Captioning Model — BART, CLAP, LLM Reranking, PyTorch Lightning, Transformers

- Built a multimodal encoder-decoder captioning system aligning audio and text in a shared embedding space for semantic audio understanding and caption generation.
- Implemented a multi-stage pipeline with diverse candidate generation (nucleus sampling), contrastive reranking with CLAP, and LLM-based refinement for final caption synthesis.

EDUCATION

New York University — New York, NY

2025 – Present

M.S., Machine Learning — Music Information Retrieval. **Sony Audio Institute Scholarship.**

Research: Multimodal representation learning with pretrained encoders for audio understanding.

Vanderbilt University — Nashville, TN

B.S., Computer Science (Engineering) — GPA 3.83 (Cum Laude, Dean's List).

2023 Blair Award for Innovative Work in Audio Technology.

SKILLS

AI & Agent Systems: Embedding/Vector Databases, Hybrid Retrieval, LangChain, LLM Agents, MCP, Ragas, RAG Pipelines

Engineering: AWS, C++, Docker, Expo, Git, HPC, JavaScript, Librosa, Node.js, Python, React, Redis, Socket.io, SQL, TypeScript